

CO₂ Assessment Test - 1

Name: S. Srinivas

Reg no: 192524129

Course code: CSA0401

Course: Operating Systems

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Analytical Problem Solving

1) A set of processes arrive at time 0 with the burst times as following:

Process	Burst time (ms)
P ₁	10
P ₂	4
P ₃	6
P ₄	2

- Calculate waiting time & ~~arrival~~ turn around time for each process using SJF (non preemptive)
- Calculate avg waiting time & turn around time.
- Interpret whether SJF improves CPU utilization compared to FCFS for the work load.

2) Round robin $\Rightarrow$ time quantum = 4 ms

Process	Burst time (ms)
P ₁	20
P ₂	5
P ₃	13
P ₄	8

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a) Draw Gantt Chart.

b) find waiting time & Turn around time

c) Analyze how changing quantum time to 10 ms affect system performance.

3) Dead lock safety checking using Banker's Algorithm.

3 resource types A, B, C $\Rightarrow$ A=10, B=5, C=7

Allocation matrix.

	A	B	C
P ₁	0	1	0
P ₂	2	0	0
P ₃	3	0	2
P ₄	2	1	1
P ₅	0	0	2

Maximum Matrix

	A	B	C
P ₁	7	5	3
P ₂	3	2	2
P ₃	4	0	2
P ₄	4	2	2
P ₅	5	3	3

a) Calculate need matrix

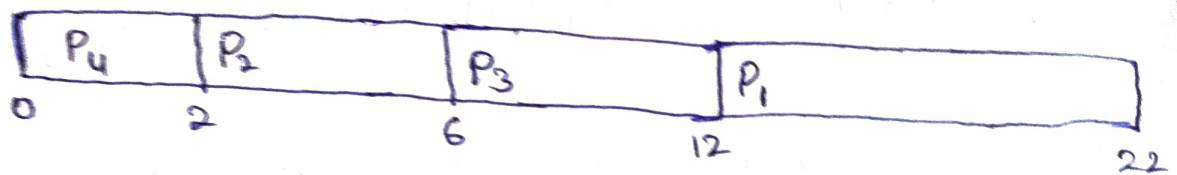
b) Determine if system is in safe state

c) Provide valid safe sequence.

1) given

Process	Burst time (ms)
P ₁	10
P ₂	4
P ₃	6
P ₄	2

2) Process Burst time
Gantt chart



a)

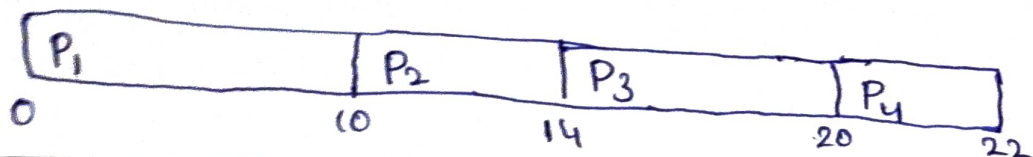
Process	Burst time	TAT = CT - AT	WT = TAT - BT
P ₁	10	22 - 10	12
P ₂	4	6 - 4	2
P ₃	6	12 - 6	6
P ₄	2	2 - 2	0

b) avg waiting time : $\frac{12 + 2 + 6 + 0}{4} = 5$

avg turn around time : $\frac{22 + 6 + 12 + 2}{4} = 10.5$

c) FCFS

Gantt Chart



Process	Burst time	TAT	WT
P ₁	10	10	0
P ₂	4	14	10
P ₃	6	20	14
P ₄	2	22	20

avg waiting time : $\frac{0 + 10 + 14 + 20}{4} = 11$

avg TAT = $\frac{10 + 14 + 20 + 22}{4} = 16.5$

The waiting time in SJF is 11 & in FCFS is 11

Waiting time in SJF is less compare with FCFS
So SJF improves CPU utilization compared to FCFS.

2) Process Burst time

P₁ 20

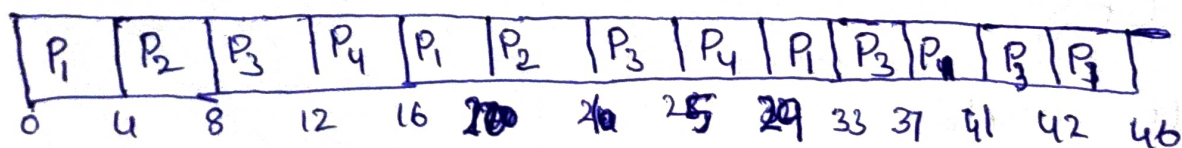
P₂ 5

P₃ 13

P₄ 8

time quantum = 4 ms

a) Gantt Chart.



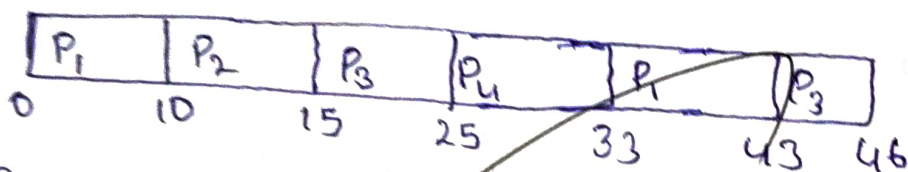
b)

Process	Burst time	TAT	WT
P ₁	20	46	26
P ₂	5	21	16
P ₃	13	42	29
P ₄	8	29	21

$$\text{Avg WT} = \frac{26 + 16 + 29 + 21}{4} = 9.13$$

$$\text{Avg Burst TAT} = \frac{46 + 21 + 42 + 29}{4} = 34.5$$

c) Gantt chart



Process	Burst time	TAT	WT
P ₁	20	43	23
P ₂	5	15	10
P ₃	13	46	33
P ₄	8	33	25

$$\text{Avg WT} = \frac{23 + 10 + 33 + 25}{4} = 22.75$$

$$\text{Avg TAT} = \frac{43 + 15 + 46 + 33}{4} = 34.25$$

The system performances becomes slow.

3) given $A = 10$, $B = 5$, $C = 7$

Available = Total - Allocated

for A : $10 - 7 = 3$

for B : $5 - 2 = 3$

for C : $7 - 5 = 2$

Available = $(3, 3, 2)$

Allocation matrix

	A	B	C
P_1	0	1	0
P_2	2	0	0
P_3	3	0	2
P_4	2	1	1
P_5	0	0	2

Maximum Matrix

	A	B	C
P_1	7	5	3
P_2	3	2	2
P_3	9	0	2
P_4	4	2	2
P_5	5	3	3

a) Need matrix :

	A	B	C
P_1	7	4	3
P_2	1	2	2
P_3	6	0	0
P_4	2	1	1
P_5	5	3	1

b) P_1 : ~~Need~~ $(7, 5, 3) > (3, 3, 2)$

P_2 : $(1, 2, 2) \leq (3, 3, 2)$

New available = $(5, 3, 2)$

P_4 : $(2, 1, 1) \leq (5, 3, 2)$

New available = $(7, 4, 3)$

P_5 : $(5, 3, 1) \leq (7, 4, 3)$

New available = $(7, 4, 5)$

P_1 : $(7, 4, 3) \leq (7, 4, 5)$

New available = $(7, 5, 5)$

P_3 : $(6, 0, 0) \leq (7, 5, 5)$, new available = $10, 5, 7$

Safe sequence

$P_2 \rightarrow P_4 \rightarrow P_5 \rightarrow P_1 \rightarrow P_3$

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4) Dead lock Detection - Resource Allocation Graph (RAG)

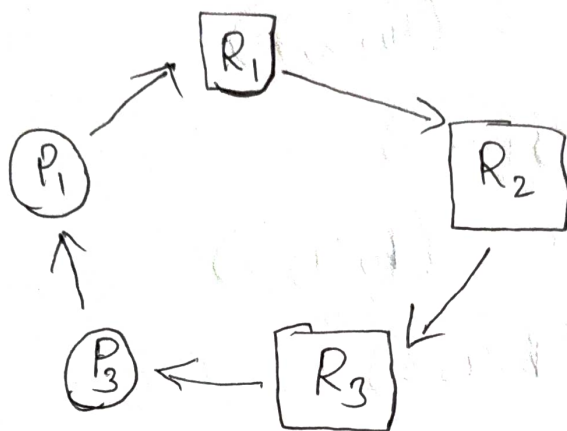
A system has following resource allocations:

- * P_1 is holding R_1 and waiting for R_2
- * P_2 is holding R_2 and waiting for R_3
- * P_3 is holding R_3 and waiting for R_1

a) Draw allocation graph.

- b) Determine whether the system is in a dead lock.
- c) Recommend a strategy to resolve it and justify the method.

a)



- b) Yes, the system is in a dead lock.

$$R_1 \rightarrow R_2 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1 \rightarrow R_1$$

c)

Recommended Strategy: Deadlock Recovery.

Justification:

Since the deadlock has already been detected,

Prevention or avoidance mechanism are no longer applicable for the current state.

5)

A System has the following Process behavior

- * Each Process spend 20% time in CPU & 80% wait for I/O
- * Degree of multiprogramming = 4 process

CPU utilization formula:

$$U = 1 - (P^n)$$

where

p = Probability process is waiting for I/O

n = no of processes in memory

a) Calculate CPU utilization

b) Compute CPU utilization if the degree of multiprogram increases to 6

c) Interpret which configuration improves system performance and why.

a) $n = 4$

$$p = 0.8$$

$$U = 1 - p^n$$

$$U = 1 - (0.8)^4$$

$$= 1 - 0.4096 = 0.5904$$

$$\text{CPU Utilization} = 59.04\%$$

b) $n = 6$

$$p = 0.8$$

$$U = 1 - (0.8)^6$$

$$= 1 - 0.262144 = 0.737856$$

$$\text{CPU Utilization} = 73.79\%$$

c) Increasing the degree of p from 4 to 6 raises the CPU utilization from 59.04% to 73.79%.